PAMANTASAN NG LUNGSOD NG MUNTINLUPA

A GENERIC POINT OF SALE SYSTEM FOR SMALL AND MEDIUM ENTERPRICES WITH DECISION SUPPORT SYSTEM

A Capstone Project Presented to the Faculty of the
College of Information Technology and Computer Studies
Pamantasan ng Lungsod ng Muntinlupa

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Bachelor of Science in Information Technology

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Chapter I
INTRODUCTION

Small and Medium Enterprises (SMEs) play an important role in supporting economic growth, employment generation, and local business development. As business operations become increasingly competitive, SMEs need efficient and reliable systems that can help manage transactions, inventory, and reports accurately. The adoption of information technology has become essential for businesses seeking to improve productivity, operational efficiency, and decision-making processes.

Despite the availability of modern business technologies, many SMEs still depend on manual methods such as handwritten records, calculators, notebooks, and spreadsheets for managing daily operations. These traditional practices often lead to computation errors, misplaced records, delayed report preparation, data redundancy, and inaccurate inventory monitoring. As a result, business owners may encounter difficulties in tracking sales performance, monitoring stock levels, and making timely business decisions.

Technology has continuously transformed the way businesses operate in today's modern economy. One of the most widely used technologies in business operations is the Point of Sale (POS) system. A POS system is designed to process sales transactions, manage inventory records, generate reports, and improve overall operational efficiency. Large companies and retail establishments commonly utilize advanced POS systems to streamline business processes and enhance customer service. However, many SMEs still lack access to affordable and efficient POS solutions that address their specific operational needs.

In addition to transaction management, business owners require access to meaningful information that can support strategic decision-making. This need has increased the importance of integrating Decision Support Systems (DSS) into business management platforms. A DSS assists users by organizing and analyzing business data to provide useful insights such as top-selling products, slow-moving items, sales summaries, low-stock alerts, and sales trends. These insights enable business owners to make informed decisions regarding inventory control, product management, and overall business performance.

To address these challenges, this study aims to develop a **Generic Point of Sale System for SMEs with Decision Support System**. The proposed system intends to automate sales transactions, inventory management, report generation, and basic business monitoring. Furthermore, it seeks to provide decision-support features that can help business owners analyze sales performance and inventory status more effectively. Through the implementation of the proposed system, SMEs may improve operational accuracy, reduce manual workload, enhance business efficiency, and support better decision-making processes.



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PROJECT CONTEXT

The study focuses on Small and Medium Enterprises (SMEs) such as retail stores, mini groceries, cafés, school supply stores, pharmacies, and similar businesses. These enterprises commonly perform daily activities involving sales transactions, inventory monitoring, and report preparation. As SMEs continue to contribute significantly to local economic development and employment generation, the need for efficient business management systems becomes increasingly important.

Many SMEs still rely on manual methods in managing business operations. Sales transactions are often recorded using handwritten receipts, notebooks, calculators, or spreadsheet applications. While these methods may be suitable for small-scale operations, they become inefficient as transaction volumes increase. Manual processes may result in inaccurate computations, misplaced records, delayed report preparation, duplicate entries, and incorrect inventory counts.

Furthermore, the absence of a centralized system makes it difficult for business owners to organize and retrieve business information efficiently. Without proper monitoring tools, SMEs may encounter challenges in tracking sales performance, monitoring inventory movement, identifying top-selling products, and maintaining accurate stock levels. These operational issues may negatively affect productivity, customer satisfaction, and business growth.

The researchers selected this study because SMEs require an affordable, practical, and user-friendly system that can support daily business operations. Although commercial POS systems are widely available, many are expensive and

contain advanced features that may not be necessary for small businesses. As a result, some SMEs continue to depend on traditional business processes despite the limitations associated with manual operations.

The researchers believe that a generic POS system integrated with basic Decision Support System (DSS) functionalities can provide SMEs with a more efficient way of managing transactions and monitoring business performance. Through organized data management and simplified analytical features, business owners can gain useful insights that support operational and managerial decision-making.

To address the identified problems, the researchers propose the development of a web-based Generic Point of Sale System for SMEs with Decision Support System. The proposed system will automate sales transaction processing, inventory management, and report generation through a centralized and user-friendly platform.

The system will include essential business management features such as product management, sales recording, inventory monitoring, receipt generation, and report preparation. In addition, it will provide basic Decision Support System functionalities including low-stock notifications, sales summaries, graphical sales reports, and identification of top-selling products.

Through the implementation of the proposed system, SMEs can reduce manual workload, improve transaction accuracy, enhance inventory monitoring, and support data-driven decision-making. The system is expected to contribute to more organized, efficient, and technology-driven business operations.



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PURPOSE AND DESCRIPTION

The purpose of this study is to develop a web-based Generic Point of Sale System with Decision Support System functionalities for Small and Medium Enterprises (SMEs). The system is designed to automate sales transactions, inventory management, report generation, and basic business monitoring.

The proposed system will help users process customer transactions, update inventory records, generate receipts, produce sales reports, and identify top-selling products. Through these features, SMEs may reduce manual workload, minimize errors, improve inventory monitoring, and enhance operational efficiency.

The system is intended to provide a practical, affordable, and user-friendly solution that supports the daily operational needs of SMEs while assisting business owners in monitoring business performance and making informed decisions.

Beneficiaries and Their Benefits

The beneficiaries of this study are the following:

1. **Small and Medium Enterprises**

   SMEs will benefit from faster sales transactions, organized inventory monitoring, and automated report generation.

2. **Business Owners**

   Business owners will benefit from sales summaries, inventory reports, top-selling product identification, and low-stock notifications that can support business decisions.

3. **Cashiers and Employees**

   Cashiers and employees will benefit from faster transaction processing, automatic computation, receipt generation, and reduced manual workload.

4. **Customers**

   Customers will benefit from faster service, accurate receipts, and improved transaction experience.

5. **Researchers and Future Developers**

   Future researchers may use this study as a reference for developing POS systems, inventory systems, and decision-support applications.

6. **IT Students**

   IT students may use this study as a learning material for system analysis, database design, software development, and business information systems.

OBJECTIVES OF THE STUDY

General Objective

The general objective of this study is to design and develop a **Generic Point of Sale System for SMEs with Decision Support System** that can automate sales transactions, inventory management, report generation, and business analytics for Small and Medium Enterprises.

Specific Objectives

The researchers specifically aim to:

1. To design a Generic Point of Sale System capable of processing sales transactions, managing inventories, generating reports, and providing decision-support functionalities.
2. To develop the proposed system using appropriate web development technologies such as PHP, MySQL, Bootstrap, JavaScript, and related software tools.
3. To test the functionality, reliability, and usability of the proposed system through Alpha Testing and Beta Testing procedures.
4. To evaluate the quality of the proposed system based on the ISO/IEC 25010:2023 Software Quality Model as assessed by IT experts.
5. To evaluate the usability and acceptability of the proposed system through actual SME users.
6. To implement the proposed system in selected Small and Medium Enterprises for operational use and evaluation.

SCOPE AND LIMITATION

This study focuses on the design and development of a web-based Generic Point of Sale System with basic Decision Support System functionalities intended for Small and Medium Enterprises (SMEs). The proposed system aims to improve and automate daily business operations by providing a centralized platform for sales transaction processing, inventory management, and report generation. The system is

specifically designed for small and medium-sized businesses such as retail stores, mini groceries, cafés, school supply stores, and similar establishments that require an organized and efficient transaction management system.

The proposed system will allow authorized users, including administrators and cashiers, to securely access and manage business operations through a centralized database system. Administrators will be responsible for managing products, monitoring inventories, viewing reports, and maintaining user accounts, while cashiers will handle sales transactions and customer purchases. The system will store transaction records and inventory information to support organized business monitoring and data management.

The proposed system includes core functionalities related to sales and inventory management. These functionalities include automated sales computation, receipt generation, transaction recording, stock updating, product inventory monitoring, and low-stock notifications. The system will also generate daily, weekly, and monthly sales reports, as well as inventory reports that may assist business owners in monitoring operational activities and evaluating business performance.

In addition, the proposed system integrates basic Decision Support System functionalities to support simple business monitoring and decision-making processes. These functionalities include the identification of top-selling products and sales summaries based on recorded transaction data. The system will also provide simple graphical reports and dashboard displays to help business owners visualize sales performance and inventory conditions.

The development of the proposed system will utilize web-based technologies and relational database management systems suitable for SME operations. Furthermore, the study includes system testing and evaluation using recognized software quality standards to assess the functionality, usability, reliability, and performance of the developed system.

This study is limited only to the development of a web-based Point of Sale System with basic Decision Support System functionalities intended for Small and Medium Enterprises. The proposed system is designed primarily for small to medium-scale business operations and is not intended for large corporations, enterprise-level organizations, or multinational retail businesses that require highly complex infrastructures and large-scale transaction processing systems.

The proposed system is limited to basic sales transaction processing, inventory management, low-stock monitoring, and report generation functionalities. The Decision Support System features are also limited only to basic sales summaries, identification of top-selling products, and simple graphical reports based on recorded transaction data. The system does not include advanced analytics, predictive forecasting, artificial intelligence technologies, machine learning algorithms, or complex business intelligence functionalities.

The study also excludes features such as online banking integration, mobile wallet payment gateways, e-commerce integration, multi-branch synchronization, cloud-based synchronization, and mobile application development. The proposed system will operate only as a web-based platform accessible through desktop or browser-enabled devices within the scope of the study.

Furthermore, the system evaluation will only involve selected respondents, IT experts, business owners, and SME users within the selected research locale. Therefore, the findings and evaluation results of the study may be limited only to the selected participants and may not fully represent all types of businesses and industries.

Chapter II

REVIEW OF THE RELATED LITERATURE AND STUDIES

This chapter presents the technical background, related literature, related studies, synthesis, and definition of terms that support the development of the proposed system. The reviewed materials provide the theoretical and technical foundation of the study by discussing Point of Sale systems, inventory management, Decision Support Systems, business reporting, and digital transformation among Small and Medium Enterprises (SMEs).

Various local and foreign sources were examined to identify existing technologies, methodologies, concepts, and practices relevant to the study. These references provide valuable insights into the design, implementation, and utilization of computerized business management systems. Furthermore, the review of related literature and studies helped identify research gaps, limitations, and opportunities that the proposed Generic Point of Sale System for SMEs with Decision Support System aims to address.

TECHNICAL BACKGROUND

The proposed Generic Point of Sale System for SMEs with Decision Support System is a web-based application designed to automate sales transactions, inventory management, report generation, and basic business monitoring. The system utilizes modern web development technologies and database management tools to ensure operational efficiency, accessibility, reliability, and maintainability.



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The front-end interface of the system was developed using HTML, CSS, Bootstrap, and JavaScript. HTML and CSS were used to create and structure the visual components of the system, while Bootstrap was utilized to develop a responsive and user-friendly interface that can function properly across different screen sizes and desktop devices. JavaScript was incorporated to provide interactive functionalities and enhance the overall user experience of the web-based application.

For back-end development, PHP was used as the primary server-side scripting language responsible for handling business logic, transaction processing, user authentication, inventory management, and report generation. PHP was selected because it is flexible, widely used, easy to deploy, and suitable for developing web-based business applications. Its compatibility with various web servers and database systems also makes it a practical choice for small and medium-sized business solutions.

The system uses MySQL as its relational database management system for storing and managing business information. MySQL was chosen because it provides structured data storage, reliable database management, and efficient retrieval of records such as product inventories, sales transactions, user accounts, and generated reports. The database ensures data consistency and supports centralized information management necessary for daily business operations.

To support the Decision Support System functionalities, the proposed system includes business monitoring features such as sales summaries, top-selling product identification, low-stock notifications, and graphical sales reports. These functionalities assist business owners in monitoring operational performance and making informed decisions based on recorded business data.

The development, coding, testing, and debugging processes were conducted using Visual Studio Code as the primary Integrated Development Environment (IDE). Visual Studio Code was selected because it provides tools for web development, debugging, code organization, and extension support, which help improve development efficiency and maintain code quality throughout the project.

The combination of these technologies provides a reliable foundation for developing a user-friendly, efficient, and maintainable Point of Sale System that meets the operational requirements of Small and Medium Enterprises.

REVIEW OF RELATED LITERATURE (RRL)

A Point-of-Sale (POS) system is described as a business management software that assists businesses in sales processing, inventory monitoring, and report generation. Studies emphasized the importance of user-friendly systems capable of reducing manual workload and improving transaction management efficiency [1]. This literature is relevant to the present study because it supports the development of a POS system that improves transaction accuracy, inventory monitoring, and overall business operations.

Integrating POS systems with Business Intelligence functionalities provides SMEs with better decision-making capabilities through data analysis and visualization. Findings showed that SMEs can gain competitive advantages by utilizing analytical tools to manage operational data efficiently [2]. This literature is relevant because the proposed system incorporates basic decision-support functionalities that help business owners monitor sales performance and inventory status.

Predictive analytics plays an important role in forecasting business demand and improving decision-making accuracy. Analytics-driven systems significantly improve forecasting performance by utilizing large volumes of business data [3]. This literature is relevant because it highlights the importance of using business data to support decision-making, although the proposed study focuses only on basic business monitoring and reporting functionalities.

The integration of decision support systems in operational management contributes to better decision-making processes by improving data organization, analysis, and interpretation capabilities [5]. This literature is relevant because the proposed system includes Decision Support System functionalities such as sales summaries, low-stock notifications, and top-selling product identification.

Inventory management systems are essential for retail companies, particularly SMEs, because they help businesses monitor inventory levels and reduce operational inefficiencies. Studies revealed that many SMEs still rely on manual inventory management practices due to the high cost of advanced systems [9]. This literature is relevant because the proposed system aims to provide an affordable solution for inventory monitoring and stock management among SMEs.

The integration of analytics functionalities into business systems improves customer engagement and operational effectiveness. Analytics-based approaches positively influence sales performance and evidence-based decision-making [10]. This literature is relevant because the proposed system provides graphical reports and sales summaries that assist business owners in evaluating business performance.

ICT adoption among SMEs continues to increase due to the growing importance of digital transformation in business operations. Studies highlighted the need for affordable and scalable information systems that can support SME development [14]. This literature is relevant because the proposed system promotes digital transformation by providing SMEs with a practical and accessible business management solution.

Web-based inventory and stock monitoring systems powered by local encrypted web servers improve real-time inventory monitoring and secure data management, which contributes to operational efficiency [15]. This literature is relevant because the proposed system is web-based and includes inventory monitoring features designed to improve data accessibility and management.

Centralized web-based stock management systems significantly improve inventory accuracy, accountability, and operational efficiency by replacing traditional spreadsheet-based workflows [16]. This literature is relevant because the proposed system utilizes a centralized database to manage sales and inventory records efficiently and accurately.

Point-of-Sale system is described as a business management software that assists businesses in sales processing, inventory monitoring, and report generation. Studies emphasized the importance of user-friendly systems capable of reducing manual workload and improving transaction management efficiency [1].

Integrating POS systems with Business Intelligence functionalities provides SMEs with better decision-making capabilities through data analysis and visualization. Findings showed that SMEs can gain competitive advantages by utilizing analytical tools to manage operational data efficiently [2].

REVIEW OF RELATED STUDIES (RRS)

A web-based POS system was developed to address stock management inaccuracies and sales discrepancies caused by manual recording systems. Findings showed that the implementation of an integrated POS system significantly improved operational efficiency and reduced inventory-related errors [17]. Similar to the present study, the system focused on transaction processing and inventory management. However, the present study also incorporates Decision Support System functionalities such as low-stock notifications, sales summaries, dashboard charts, and top-selling product identification.

A web-based POS system designed for a food business using the Waterfall development methodology demonstrated improvements in transaction accuracy, customer experience, and operational efficiency [18]. Similar to the present study, the system automated sales transactions and improved operational processes. However, the proposed study extends its functionality by providing decision-support features that assist business owners in monitoring sales and inventory performance.

A website-based POS system developed for a culinary business addressed transaction processing and financial reporting issues caused by manual operations. The study revealed improvements in transaction handling and report generation efficiency [19]. Similar to the present study, the system focused on transaction recording and reporting. However, the present study additionally provides inventory monitoring and business monitoring functionalities through graphical reports and sales summaries.

A web-based POS system developed as a financial digitalization solution for MSMEs reduced transaction recording errors and improved financial transparency through centralized data management [20]. Similar to the present study, the system promotes digital transformation and centralized data management. However, the proposed study also includes decision-support functionalities that help business owners monitor product performance and inventory status.

A web-based POS system designed for transaction recording and inventory monitoring in SMEs included role-based access functionalities for administrators, owners, and cashiers to improve business management and operational efficiency [21]. Similar to the present study, the system provides inventory monitoring and role-based access control. However, the present study integrates additional business monitoring features such as low-stock alerts, sales summaries, and top-selling product identification.

Studies on implementing web-based POS systems for SMEs concluded that these systems are feasible and beneficial despite technological adoption challenges in developing countries [22]. Similar to the present study, the reviewed systems were developed to improve SME operations through digital solutions. However, the proposed study focuses on combining POS functionalities with basic Decision Support System features to further support business decision-making.

The integration of Rapid Application Development (RAD) and Design Thinking methodologies in developing web-based POS and inventory management systems resulted in high user acceptance and operational effectiveness [23]. Similar to the present study, the research utilized a structured development approach to ensure system usability and effectiveness. However, the proposed study incorporates additional decision-support functionalities for business monitoring.

The implementation of web-based transaction and stock management systems improved transaction efficiency and inventory accuracy in retail operations [24]. Similar to the present study, the system emphasized transaction processing and inventory management. However, the present study also provides sales summaries, dashboard charts, and low-stock notifications for improved business monitoring.

Web-based POS systems developed using the prototyping method emphasized the importance of user involvement and iterative system development in producing user-friendly business applications [25]. Similar to the present study, the research focused on developing a user-friendly POS system. However, the proposed study expands its capabilities by integrating Decision Support System functionalities that assist business owners in evaluating business performance.

Integrated POS system architectures for MSMEs highlighted the importance of scalable and integrated systems in supporting digital transformation among SMEs [26]. Similar to the present study, the system supports SME digitalization through integrated business management tools. However, the proposed study specifically focuses on practical and affordable functionalities suitable for small and medium-sized enterprises.

Locally developed web-based ordering systems demonstrated the advantages of integrating ordering systems and business reporting functionalities to improve customer experience and operational management [27]. Similar to the present study, the system includes reporting functionalities that support business operations. However, the proposed study additionally incorporates inventory monitoring and decision-support features.

Studies on cloud-based POS systems for inventory management among micro food businesses emphasized that system quality and usability significantly affect technology adoption in SMEs [28]. Similar to the present study, the research highlights the importance of system usability and operational efficiency. However, the proposed study focuses on a web-based platform with basic Decision Support System functionalities rather than cloud-dependent operations.

Point-of-Sale systems integrated with CRM and appointment scheduling functionalities improved inventory management, customer relationship management, and sales monitoring [29]. Similar to the present study, the system supports sales and inventory management. However, CRM and appointment scheduling are outside the scope of the present study, which focuses on essential POS and decision-support functionalities for SMEs.

Automated POS systems focused on profitability measurement and stock monitoring improved operational efficiency and inventory monitoring effectiveness in retail businesses [30]. Similar to the present study, the system supports inventory monitoring and operational efficiency. However, the proposed study emphasizes practical business monitoring features such as low-stock notifications, sales summaries, dashboard charts, and top-selling product identification.

SYNTHESIS

The reviewed literature and studies emphasized the importance of Point-of-Sale systems, inventory management systems, Decision Support Systems, and digital business technologies in improving operational efficiency and supporting decision-making among Small and Medium Enterprises (SMEs). Previous studies consistently showed that manual transaction processing and inventory management often result in inaccurate records, delayed reporting, inventory discrepancies, and operational inefficiencies.

Most of the reviewed systems focused on sales transaction processing, inventory monitoring, and report generation. Several studies also incorporated analytical and reporting functionalities to improve business monitoring and operational management. These systems demonstrated the benefits of automation in reducing manual workload, improving data accuracy, and enhancing overall business performance.

Although previous studies developed Point-of-Sale systems for transaction processing and inventory management, many of them have limited decision-support features specifically designed for SMEs. This shows a research gap in developing an affordable and user-friendly POS system that combines sales processing, inventory monitoring, report generation, and basic Decision Support System functionalities for small and medium enterprises.

Furthermore, the reviewed literature revealed that many SMEs continue to experience challenges in adopting digital technologies due to financial constraints, limited technical expertise, and the complexity of some existing business management systems. These challenges highlight the need for practical and accessible solutions that can support both operational management and business monitoring.

Based on the findings of the reviewed literature and studies, the researchers propose the

development of a web-based Generic Point of Sale System with Decision Support System functionalities. The proposed system aims to automate sales transactions, inventory management, and report generation while providing business monitoring features such as low-stock notifications, sales summaries, graphical sales reports, and top-selling product identification. Through these functionalities, the system seeks to improve operational efficiency, support informed decision-making, and provide an affordable business management solution for SMEs.

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DEFINITION OF TERMS

Bootstrap

Refers to a front-end web development framework used to create responsive and user-friendly interfaces for the proposed system.

Business Analytics

Refers to the process of examining sales records, inventory data, and business reports to support monitoring and decision-making within the proposed system.

Business Owner

Refers to the individual responsible for managing and overseeing business operations, sales, inventory, and reports using the proposed system.

Cashier

Refers to the authorized user responsible for processing customer transactions, recording sales, and generating receipts through the Point-of-Sale system.

CSS (Cascading Style Sheets)

Refers to a stylesheet language used to control the visual appearance, layout, and design of web pages in the proposed system.

Database Management System (DBMS)

Refers to a software system used to store, organize, retrieve, and manage structured data efficiently within the proposed application.

Decision Support System (DSS)

Refers to the functionality integrated into the proposed system that assists business owners in monitoring sales and inventory performance through sales summaries, low-stock notifications, top-selling product identification, and graphical reports.

HTML (HyperText Markup Language)

Refers to the standard markup language used to create and structure the web pages of the proposed system.

Inventory Management

Refers to the process of monitoring, organizing, updating, and controlling product stock levels and inventory records within the proposed system.

JavaScript

Refers to a scripting language used to create interactive functionalities and dynamic components within the proposed web-based system.

MySQL

Refers to the relational database management system utilized for storing and managing sales records, inventory information, user accounts, and transaction data.

PHP (Hypertext Preprocessor)

Refers to the server-side scripting language used to develop the business logic and functionalities of the proposed system.

Point of Sale (POS) System

Refers to the proposed web-based system designed to automate sales transactions, inventory management, report generation, and business monitoring for Small and Medium Enterprises.

Sales Analytics

Refers to the feature of the proposed system that summarizes and analyzes transaction records to identify top-selling products and provide sales performance information.

Small and Medium Enterprises (SMEs)

Refers to small and medium-scale businesses such as retail stores, mini groceries, cafés, school supply stores, and similar establishments that are the intended users and beneficiaries of the proposed system.

User Authentication

Refers to the process of verifying the identity of authorized users through usernames and passwords before granting access to the system.

Visual Studio Code

Refers to the Integrated Development Environment (IDE) used by the researchers for coding, debugging, testing, and developing the proposed system.

Web-Based System

Refers to a software application that can be accessed through a web browser without requiring standalone installation on user devices.

refers to the process of analyzing sales records, inventory data, and transaction reports to identify business trends, monitor performance, and support decision-making within the proposed system.

Chapter III

METHODOLOGY

This chapter presents the methodology used in the development of the proposed system. It discusses the procedures and methods used by the researchers in analyzing requirements, documenting system features, designing the system architecture, developing and testing the system, and planning its implementation. The chapter also includes diagrams and process descriptions that explain how the proposed system will function.

REQUIREMENT ANALYSIS

Requirements analysis is the process of identifying and understanding the needs of the intended users of the proposed system. In this study, the researchers analyzed the existing business process commonly used by Small and Medium Enterprises (SMEs) in handling sales transactions, inventory monitoring, and report generation. This phase helped the researchers determine the problems in the current process and identify the necessary features that must be included in the proposed system.

The researchers considered the following elements in analyzing the requirements of the system:

Who refers to the people involved in the system. The main users of the proposed system are the administrator, business owner, and cashier. The administrator is responsible for managing products, users, inventory records, and reports. The cashier is responsible for processing sales transactions and issuing receipts. The business owner monitors sales, inventory, and business reports.

What refers to the business activities involved in the study. These include sales transaction processing, product inventory management, stock monitoring, report generation, and basic decision support through sales summaries, low-stock notifications, and top-selling product identification.

Where refers to the environment where the system will be used. The proposed system is intended for Small and Medium Enterprises such as retail stores, mini groceries, cafés, school supply stores, and similar small business establishments.

When refers to the timing of the business process. The system will be used during daily business operations, particularly during customer transactions, inventory updates, sales monitoring, and report preparation.

How refers to how the current procedures are performed. In many SMEs, sales transactions are manually recorded using notebooks, receipts, calculators, or spreadsheet files. Inventory is commonly monitored through manual counting, while sales reports are prepared by reviewing handwritten records or spreadsheet entries. These procedures are time-consuming and prone to errors.

Existing System Procedure

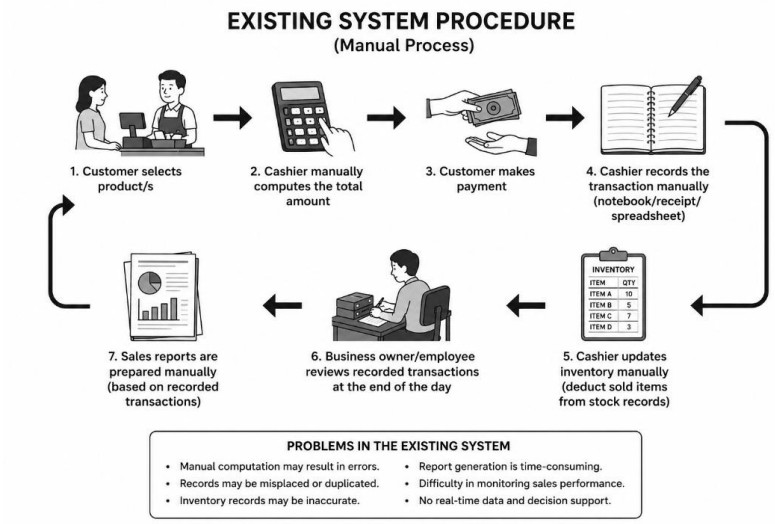


Figure 1. Manual Process Existing System

In the current manual process, the cashier receives the customer's order or purchased items and manually computes the total amount. After payment is received, the transaction is recorded in a notebook, receipt, or spreadsheet. Inventory updates are also performed manually by deducting sold items from stock records. At the end of the day, the business owner or employee prepares sales reports by checking the recorded transactions. This process may lead to inaccurate computations, delayed reporting, missing records, and inconsistent inventory monitoring.

Existing System Flowchart

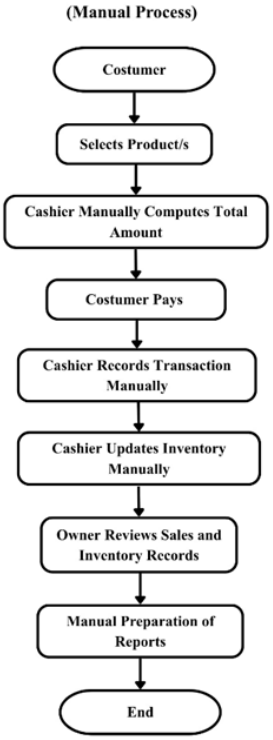


Figure 2. Manual Process Flowchart Existing System

Problems Identified in the Existing System

The existing manual process presents several problems. First, manual computation may result in incorrect totals and change calculation. Second, handwritten or spreadsheet-based records may be misplaced, duplicated, or incorrectly encoded. Third, inventory records may not be updated immediately after each transaction, resulting in inaccurate stock monitoring. Fourth, preparing daily, weekly, and monthly reports requires additional time and effort. Lastly, business owners may have difficulty identifying top-selling products and low-stock items without an organized system.

REQUIREMENT DOCUMENTATION

Requirement documentation establishes the basis for agreement between the users and the developers regarding what the proposed system should accomplish. This section presents the major modules and features of the proposed system.

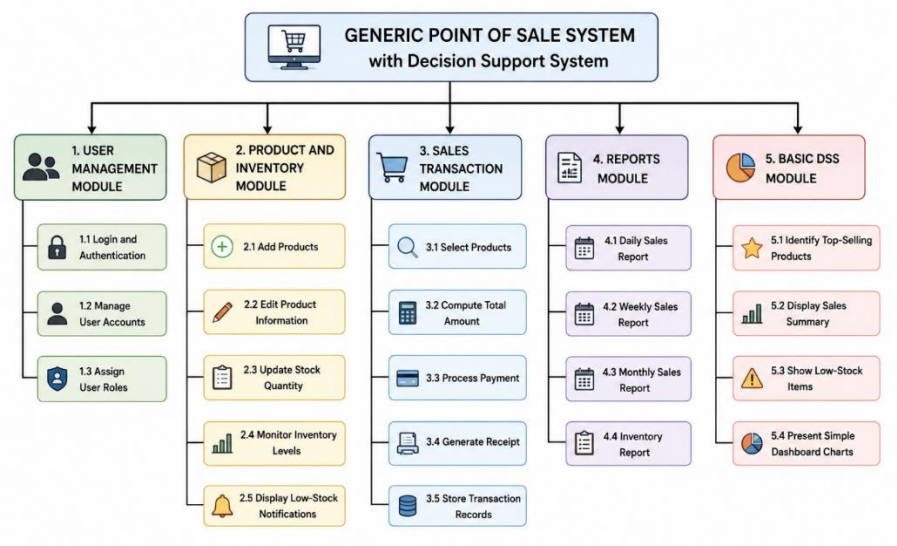


Figure 3. Functional Decomposition Diagram

The figure presents the Functional Decomposition Diagram of the proposed Generic Point of Sale System with Decision Support System. It illustrates the major modules of the system and their corresponding functions. The system is composed of five primary modules: User Management, Product and Inventory Management, Sales Transaction, Reports, and Basic DSS Module. Each module is further divided into specific functions such as user authentication, inventory monitoring, sales processing, report generation, and decision support features. This diagram provides a hierarchical view of the system's structure and shows how each module contributes to the overall functionality of the proposed system.

User Management Module

The User Management Module allows authorized users to access the system based on their assigned roles. The administrator can create, update, and manage user accounts. This module ensures that only authorized users can access sensitive information such as sales records, inventory records, and reports.

Product and Inventory Module

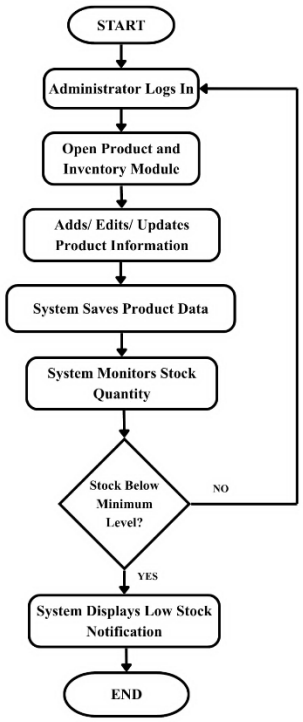


Figure 4. Product and Inventory Module Process

The Product and Inventory Module allows the administrator to add, edit, delete, and update product information. It also monitors stock quantities and displays low-stock notifications when an item reaches its minimum stock level. This module helps SMEs maintain accurate product records and prevent stock shortages.

Sales Transaction Module

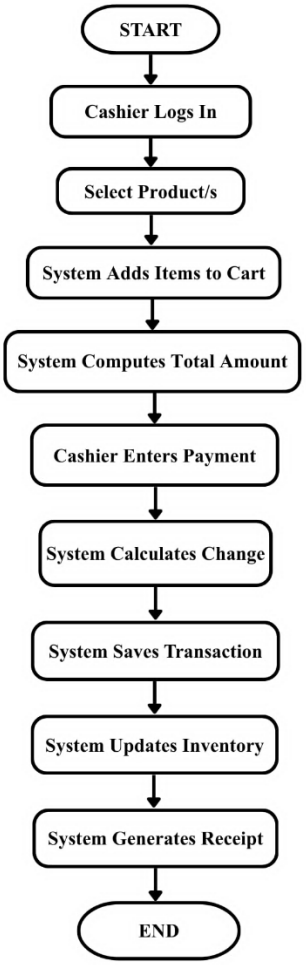


Figure 5. Sales Transaction Process

The Sales Transaction Module allows the cashier to process customer purchases efficiently. It computes the total amount automatically, records payment details, calculates change, updates inventory after each transaction, and generates receipts. This module reduces manual computation errors and improves transaction speed.

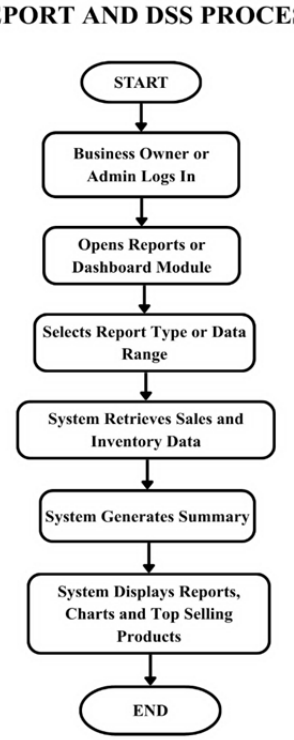


Figure 6. Report and DSS Process

Reports Module

The Reports Module generates daily, weekly, and monthly sales reports. It also provides inventory reports that summarize product availability and stock movement. These reports help business owners monitor business performance and make operational decisions.

Basic Decision Support System Module

The Basic DSS Module provides simple decision support features such as identifying top-selling products, displaying sales summaries, showing low-stock items, and presenting graphical dashboard reports. These features assist business owners in monitoring sales performance and inventory conditions without using advanced artificial intelligence or predictive forecasting.

SYSTEM ARCHITECTURE

The proposed system follows a web-based client-server architecture. Users will access the system through a web browser using a desktop or browser-enabled device. The web interface will communicate with the application server, which processes business logic such as login validation, sales transactions, inventory updates, and report generation. The database server stores user accounts, product records, transaction records, inventory data, and sales reports.

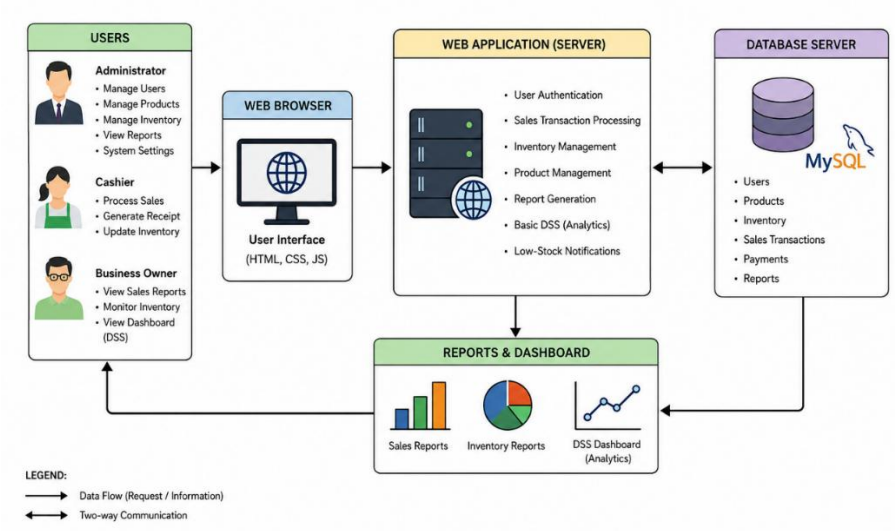


Figure 7. System Architecture Diagram

The user interacts with the system through the web browser. The browser displays the user interface for login, product management, sales transactions, inventory monitoring, reports, and dashboard features. The web application handles the processing of user requests, such as saving sales transactions, updating stock levels, generating reports, and displaying decision support information. The MySQL database stores and manages all business records in a structured and centralized manner.

DESIGN OF SOFTWARE, SYSTEMS, PRODUCT AND/OR PROCESSES

This section presents the design of the proposed system, including the conceptual framework, proposed system process, and data flow. The system is designed to provide a practical computing solution for SMEs by automating manual transaction and inventory processes.

Conceptual Framework

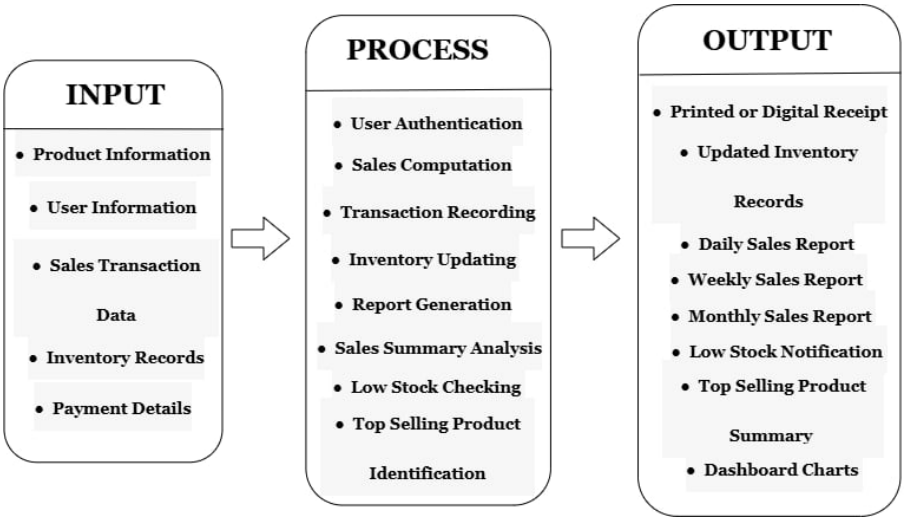


Figure 8. Conceptual Framework

The conceptual framework of the study was based on the Input–Process–Output (IPO) model, which provides a structured approach in identifying the necessary components, procedures, and expected results in the development of the proposed system. The framework illustrates how the collected data and system requirements are processed to produce useful outputs that support the operations and decision-making activities of Small and Medium Enterprises (SMEs).

In the Input phase, essential information and resources required for the development and operation of the system were identified. These inputs include product information, user information, sales transaction data, inventory records, and payment details. Product information contains details such as product names, prices, categories, and available stock quantities, while user information includes authorized accounts such as administrators and cashiers. Sales transaction data, inventory records, and payment details serve as the primary data sources used by the system for processing transactions, monitoring stocks, and generating accurate business reports.

In the Process phase, the proposed system performs the necessary operations to transform the collected inputs into meaningful information. This phase includes user authentication to ensure secured system access, automated sales computation for accurate transaction processing, transaction recording for maintaining sales history, and inventory updating to monitor product availability. The system also performs report generation, sales summary analysis, low-stock checking, and top-selling product identification as part of its basic Decision Support System functionalities. These processes allow business owners to analyze important business information and support operational decision-making.

The Output phase represents the results generated by the system after processing the provided data. The proposed system produces printed or digital receipts, updated inventory records, daily, weekly, and monthly sales reports, low-stock notifications, top-selling product summaries, and dashboard charts. These outputs provide SMEs

with organized and accessible business information that can help improve transaction management, inventory monitoring, and overall business operations.

Proposed System Flowchart

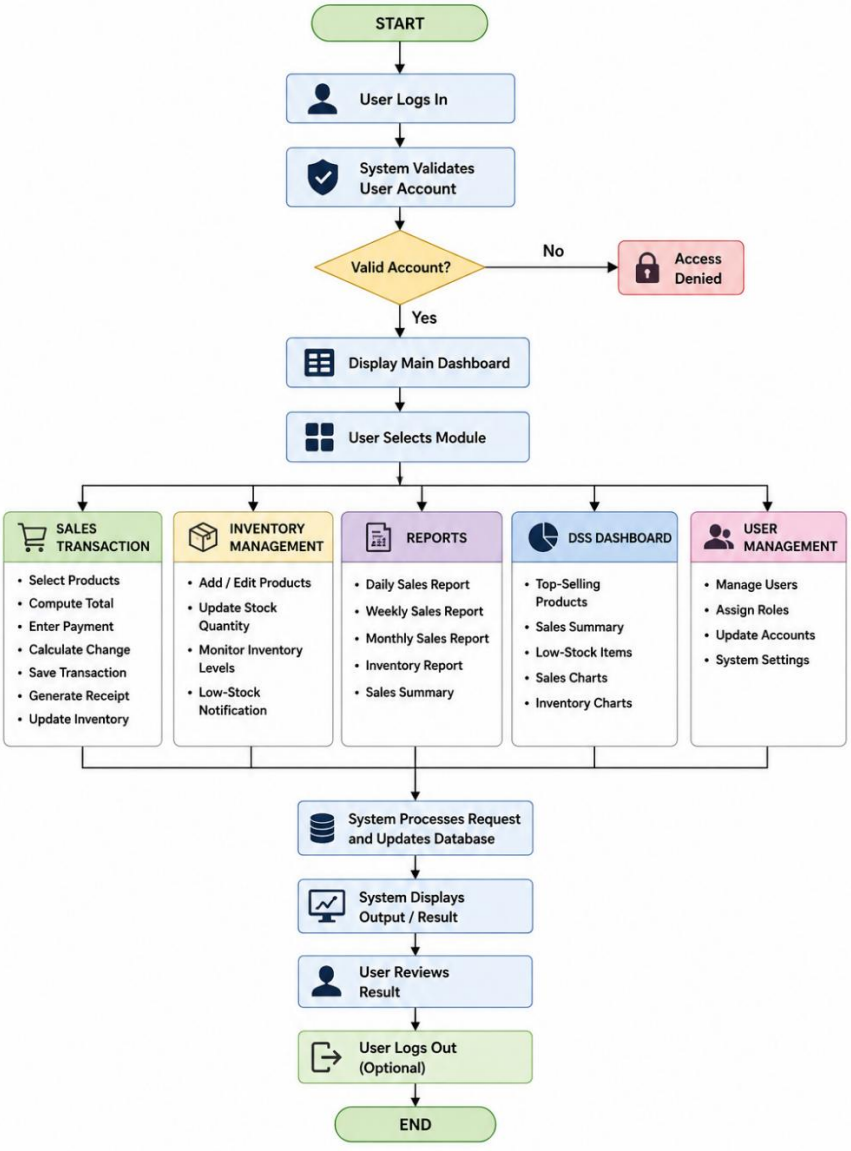


Figure 9. System Flowchart

The figure illustrates the overall process flow of the proposed Generic Point of Sale System with Decision Support System. The process begins when the user logs into the system and the system validates the user account credentials. If the account is invalid, access is denied; otherwise, the user is directed to the main dashboard. From the dashboard, the user can access different system modules, including Sales Transaction, Inventory Management, Reports, DSS Dashboard, and User Management. Each module performs specific functions such as processing sales, monitoring inventory, generating reports, identifying top-selling products, and managing user accounts. After a request is submitted, the system processes the information, updates the database, and displays the corresponding output or result. The user can then review the generated information and optionally log out of the system, completing the transaction cycle. This flow ensures efficient transaction processing, accurate inventory monitoring, organized report generation, and basic decision support for business operations.

DEVELOPMENT AND TESTING

The researchers will use the Rapid Application Development (RAD) methodology in developing the proposed system. RAD was selected because it supports faster development, user feedback, and continuous improvement. This methodology is suitable for the proposed system because the project requires user interface design, system prototyping, module development, testing, and refinement based on user evaluation.

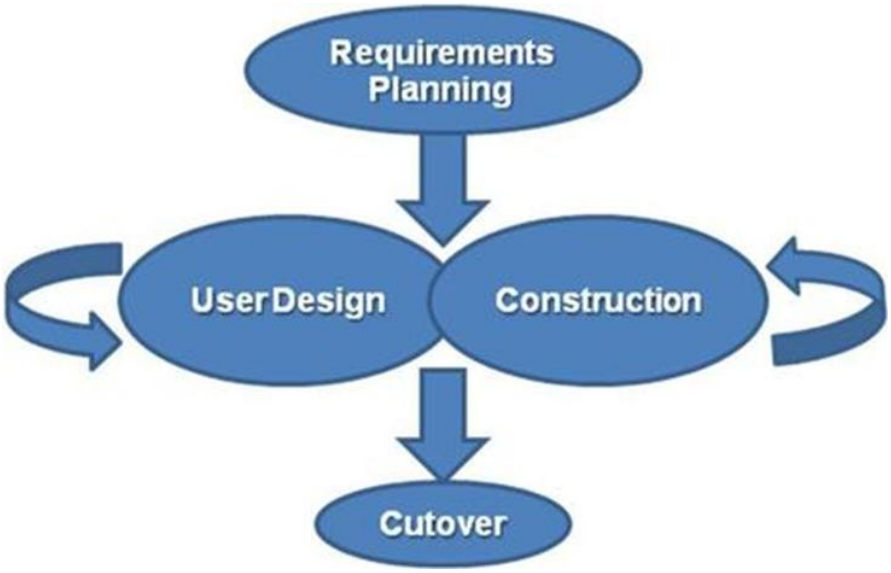


Figure 10. RAD Methodology Diagram

Phase 1: Requirement Planning

In this phase, the researchers will gather the necessary information about the current sales and inventory management process used by SMEs. The researchers will identify the problems encountered in manual transaction processing, inventory monitoring, and report generation. The system requirements, user roles, needed modules, and limitations of the proposed system will also be determined during this phase.

Phase 2: User Design

In this phase, the researchers will design the user interface, system flow, database structure, and module layout of the proposed system. The researchers will prepare diagrams such as system architecture, flowcharts, and functional decomposition diagrams. Feedback from potential users may also be considered to improve the usability and practicality of the system design.

Phase 3: Construction

In this phase, the actual development of the system will be conducted. The researchers will develop the frontend interface, backend processes, and database structure. The major modules to be developed include User Management, Product and Inventory Management, Sales Transaction, Reports, and Basic DSS Dashboard. The system will be coded, connected to the database, and tested during development to ensure proper functionality.

Phase 4: Cutover

In this phase, the completed system will be prepared for final testing, evaluation, and implementation. The researchers will conduct system testing, correct identified errors, finalize system features, and prepare the system for deployment. User orientation and demonstration may also be conducted to help users understand how to operate the system.

Testing Procedure

The proposed system will undergo testing to determine whether all modules function according to the intended requirements. Testing will help identify errors, verify system outputs, and evaluate the usability and reliability of the system.

Alpha Testing

No.	Module / Feature	Objectives	Test Activities	Testers	Expected Result	Output
1	Login and User Authentication	To verify that users can securely log in and access the system based on their roles.	• Test valid and invalid login • Test role-based access • Test logout process	Researchers and Selected IT Experts	Users can log in successfully and access only authorized modules.	Pass / Fail
2	Product Management	To ensure product information can be added, updated, deleted, and viewed correctly.	• Add new product • Edit product information • Delete product • View product list	Researchers and Selected IT Experts	Product data is saved, updated, deleted, and displayed accurately.	Pass / Fail
3	Inventory Management	To verify that inventory levels are updated and monitored accurately.	• Update stock quantity • Check inventory computation • Monitor inventory levels	Researchers and Selected IT Experts	Inventory quantities are updated correctly and displayed properly.	Pass / Fail
4	Sales Transaction Processing	To ensure sales transactions are processed correctly.	• Select products • Compute total amount • Process payment	Researchers and Selected IT Experts	Transactions are processed correctly with accurate totals and payment records.	Pass / Fail
5	Receipt Generation	To verify that receipts are generated accurately.	• Generate receipt after sale • Check receipt details • Print / preview receipt	Researchers and Selected IT Experts	Receipts are generated with correct transaction information.	Pass / Fail
6	Report Generation	To ensure reports are generated correctly based on data.	• Generate daily report • Generate weekly report • Generate monthly report • Generate inventory report	Researchers and Selected IT Experts	Reports are accurate and match the recorded transactions.	Pass / Fail
7	Low-Stock Notification	To check if the system notifies users of low-stock items.	• Set minimum stock level • Reduce stock quantity • Check notification alert	Researchers and Selected IT Experts	Low-stock items are detected and notifications are displayed correctly.	Pass / Fail
8	Basic DSS Dashboard	To verify dashboard analytics and summaries.	• Check sales summary • Check top-selling products • Check dashboard charts	Researchers and Selected IT Experts	Dashboard displays accurate analytics and visuals.	Pass / Fail

Table 1. Alpha Testing Plan

The table presents the Alpha Testing Matrix of the proposed Generic Point of Sale System with Decision Support System. It outlines the system modules to be tested, the objectives of each test, the testing activities to be performed, the assigned testers, the expected results, and the corresponding outputs. Alpha testing will be conducted by the researchers and selected IT experts to verify the functionality, accuracy, and reliability of each system module before deployment. The results of this testing phase will help identify and correct system errors, ensuring that all features operate according to the specified requirements.

Alpha testing will be conducted by the researchers and selected IT experts. This test will focus on checking the functionality of each module, identifying system errors, verifying database operations, and ensuring that the system performs according to its requirements.

Beta Testing

No.	Aspect	Objectives	Participants	Test Activities	Evaluation Method	Expected Result	Output
1	Usability Testing	To determine if the system is easy to understand and use.	• Business Owners • Cashiers • SME Users	• Perform tasks in each module • Observe user interaction • Identify usability issues	Survey / Questionnaire (ISO/IEC 25010:2011)	Users can navigate and use the system easily with minimal assistance.	Accept / Revise
2	Functionality Testing	To validate that the system functions as expected in real business operations.	• Business Owners • Cashiers • SME Users	• Process actual sales • Manage products and inventory • Generate reports • Check system data accuracy	Survey / Questionnaire (ISO/IEC 25010:2011)	All modules work correctly and support daily business operations.	Accept / Revise
3	Acceptability Testing	To determine if the system meets the needs and expectations of users.	• Business Owners • Cashiers • SME Users	• Evaluate overall system • Provide feedback and suggestions • Rate system performance	Survey / Questionnaire (ISO/IEC 25010:2011)	The system is acceptable, practical, and applicable to business operations.	Accept / Revise

Table 2. Beta Testing Plan

User Type	Role in Beta Testing
Business Owners	• Evaluate if the system supports their business operation • Provide suggestions for improvement
Cashiers	• Test daily transaction Process • Evaluate ease of use in processing sales in printing receipts
SME Users	• Tests system features and reports • Provide feedback based on actual usage

Table 3. Participants in Beta Testing

Step	Description
1.Select Participants	Identify and invite actual users to participate in beta testing
2.Provide Orientation	Explain the system features, functions, and tasks to the participants
3.Perform Testing	Users perform tasks and use the system in real scenarios
Collect Feedback	Gather feedback, issues, and suggestions from the participants

Table 4. Beta Testing Flow

Beta testing will be conducted with selected actual users such as business owners, cashiers, and SME users. This test will determine whether the system is understandable, usable, and applicable to actual business operations. Feedback from users will be used to improve the system.

Evaluation Instrument

The proposed system will be evaluated using the ISO/IEC 25010:2011 Software Quality Model. The evaluation will focus on the following criteria:

No.	Criteria	Description	Indicators / Questions	Rating Scale 1	2	3	4	5
1	Functional Suitability	The degree to which the system provides functions that meet stated and implied needs.	• The system functions meet user requirements. • All functions are available and work as intended. • The system produces accurate results. • The system is appropriate for the tasks.					
2	Performance Efficiency	The performance relative to the amount of resources used under stated conditions.	• The system responds quickly to user actions. • The system can handle many transactions efficiently. • Pages load fast. • The system uses resources efficiently.					
3	Compatibility	The degree to which the system can exchange information with other systems and perform its required functions in its environment.	• The system works well with other systems. • The system works on different browsers. • The system is compatible with the environment. • Data can be imported/exported correctly.					
4	Usability	The degree to which the system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction.	• The system is easy to learn. • The system is easy to navigate. • The interface is clear and understandable. • The system is user-friendly.					
5	Reliability	The degree to which the system performs specified functions under stated conditions for a specified period of time.	• The system is stable and error-free. • The system works as expected. • The system recovers from errors. • The system is available when needed.					
6	Security	The degree to which the system protects information and data so that unauthorized people cannot access it.	• User accounts are secured. • Only authorized users can access the system. • Data is protected from unauthorized access. • The system has proper access control.					
7	Maintainability	The degree to which the system can be modified effectively and efficiently.	• The system is easy to maintain. • The code is well-structured. • The system can be easily updated. • Bugs can be fixed easily.					
8	Portability	The degree to which the system can be transferred from one environment to another.	• The system can be deployed in another environment. • The system can run on different devices/platforms. • The system has minimal environment dependencies. • The system is easy to install and configure.					

RATING SCALE

1 - Very Poor	2 - Poor	3 - Fair	4 - Good	5 - Excellent
The criterion is not satisfied.	The criterion is rarely satisfied.	The criterion is moderately satisfied.	The criterion is mostly satisfied.	The criterion is fully satisfied.

Figure 11. ISO/IEC 25010:2011 Software Quality Evaluation Instrument

The figure presents the ISO/IEC 25010:2011 Software Quality Evaluation Instrument that will be used to assess the quality of the proposed system. It includes eight software quality criteria: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Respondents will evaluate the system using a five-point rating scale to determine its overall functionality, usability, reliability, and acceptability.

Statistical Treatment

The researchers may use weighted mean to interpret the results of the system evaluation. The responses from IT experts and actual users will be analyzed to determine the overall acceptability and quality of the proposed system.

IMPLEMENTATION PLAN

The implementation plan describes how the proposed system will be deployed and introduced to the intended users. The system will be implemented as a web-based application accessible through desktop or browser-enabled devices. The implementation will involve system installation, database setup, user account creation, testing, user orientation, and actual system demonstration.

Implementation Activities

Activity 1: Prepare System Files and Database

Table Name	Description
users	Stores user accounts and roles
products	Stores product information
categories	Stores product categories
inventory	Tracks stock quantity and levels
sales	Stores sales transaction header
sales_items	Stores items included in a sale
payments	Stores payment details
reports	Stores generated reports (optional)

Table 5. Database Tables Description

The researchers will prepare the final system files and configure the database for storing user accounts, products, inventory records, transactions, and reports.

Activity 2: Install and Configure the System

The system will be installed on a local server or hosting environment depending on the available resources of the study.

Activity 3: Create User Accounts

Features	Administrator	Cashier	Business Owner
User Management	✅	❌	❌
Product Management	✅	❌	✅
Inventory Management	✅	✅	✅
Sales Transction	✅	✅	❌
Reports	✅	✅	✅
Dashboard/DSS	✅	✅	✅
System Settings	✅	❌	❌

Table 6. User Access and Permissions Matrix

Administrator and cashier accounts will be created to allow authorized users to access the system. Each account will be assigned specific roles and permissions based on the responsibilities of the user. This ensures that sensitive information and system functions are only accessible to authorized personnel, thereby improving security, data integrity, and proper system management.

Activity 4: Encode Initial Product Data

Initial product information such as product name, price, quantity, and minimum stock level will be encoded into the system.

Activity 5: Conduct System Testing

The researchers will test the system modules to ensure that sales transactions, inventory updates, reports, and dashboard features are working properly.

Activity 6: Conduct User Orientation

The researchers will provide a short orientation or demonstration to selected users on how to operate the system.

Activity 7: Deploy the System for Evaluation

The system will be used by selected respondents for evaluation and feedback gathering.

Required Resources

The development and implementation of the proposed system require the following resources in terms of hardware, software, and human resources

Resource Category	Requirements	Purpose / Description
Hardware Requirements	- Desktop computer or laptop - Internet connection or local network - Optional receipt printer - Optional barcode scanner	These hardware components are necessary for running the system, accessing the application, and processing transactions. The optional devices enhance the efficiency of sales and inventory operations.
Software Requirements	- Web browser - Web server environment - MySQL database - Code editor - PHP or selected backend technology	These software components are required for developing, running, and managing the system. They support the system's functionality, database management, and web-based operations.
Human Resources	- Researchers / Developers - IT Experts - Business Owner - Cashier / End Users	These individuals are involved in the development, evaluation, and daily use of the system. Their roles ensure the successful implementation and continuous improvement of the proposed system.

Table 7. Required Resources

The table presents the hardware, software, and human resources required for the development, deployment, and operation of the proposed system. It identifies the necessary equipment, development tools, and personnel needed to support the implementation process. These resources ensure that the system can be developed efficiently, tested properly, and utilized effectively by its intended users.

Phase	Time Frame	Major Tasks / Activities	Description	Resources Needed	Person Responsible	Expected Output
1	Week 1 – 2	Finalization of System Requirements and Design	Gather and finalize system requirements, analyze business processes, and design system architecture, database, and user interface.	- Computer - Software tools (e.g., Draw.io, MS Word) - Internet	Researchers / Developers IT Experts Business Owner	Approved requirements document and system design.
2	Week 3 – 4	Database Creation and Module Setup	Create database structure, set up the development environment, and prepare the basic modules.	- Computer - Web Server (XAMPP) - MySQL - Code Editor - Internet	Developers IT Experts	Database created and initial modules set up.
3	Week 5 – 6	Development of Sales Transaction and Inventory Modules	Develop modules for product management, sales transaction processing, payment handling, and inventory management.	- Computer - Development tools - Database - Internet	Developers	Completed sales transaction and inventory modules.
4	Week 7 – 8	Development of Report and DSS Dashboard Modules	Develop modules for report generation (daily, weekly, monthly), low-stock notification, top-selling product summary, and dashboard.	- Computer - Development tools - Database - Internet	Developers IT Experts	Completed report and DSS dashboard modules.
5	Week 9 – 10	System Testing and Debugging	Test all system modules, identify and fix errors, and ensure the system functions as intended.	- Computer - Test Data - Internet	Developers IT Experts	Tested and debugged system.
6	Week 11 – 12	User Evaluation and Implementation	Conduct user evaluation, gather feedback, provide training, and deploy the system for actual use.	- Computer - Internet - Training Materials	Researchers Business Owner Cashier / End Users	System implemented and accepted by users.
7	Week 13 – 15	Revision and Final Documentation	Revise the system based on feedback and prepare the final documentation.	- Computer - Documentation Tools (e.g., MS Word)	Researchers / Developers	Final system and complete documentation.

Table 8. Implementation Schedule Plan

The table presents the implementation schedule of the proposed Generic Point of Sale System with Decision Support System. It outlines the major activities, time frames, required resources, responsible personnel, and expected outputs throughout the development and implementation phases. The schedule serves as a guide for managing project tasks and monitoring progress to ensure the successful completion and deployment of the system within the planned timeframe.

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